P-R-1272C
March 19, 1976
SUPERSEDING
P-R-1272B
April 7, 1972

FEDERAL SPECIFICATION

RINSE ADDITIVE, DISHWASHING, CONCENTRATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a low-foaming liquid rinse additive concentrate for use in the final rinse cycle of a mechanical dishwashing machine equipped with a suitable rinse injector device (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specifications:

MIL-B-26701 - Bottles, Screw Cap, and Carboys, Polyethylene Plastic.
MIL-I-43728 - Injector, Rinse, Automatic, for Commercial Dishwashing Machines.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 482 - Ash of Petroleum Products.
D 1331 - Surface and Interfacial Tension of Solutions of Surface-Active Agents.
D 2024 - Cloud Point of Nonionic Surfactants.
D 2834 - Nonvolatile Matter (Total Solids) in Water Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer Emulsion Floor Polishes.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Soap and Detergent Association:

Test Procedure and Standards - ABS and LAS Biodegradability Scientific and Technical Report No. 3.
Status of Biodegradability Testing of Nonionic Surfactant Scientific and Technical Report No. 6.

(Application for copies should be addressed to the Soap and Detergent Association, 475 Park Avenue South, New York, NY 10016.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The rinse additive shall consist of wetting agents, combined with or without a defoamer, as necessary, to meet the requirements of section 3. The rinse additive shall be suitable for use in a mechanically activated or water-powered injector mechanism.

3.2 Excluded ingredients. The rinse additive shall contain no ethylene glycol when tested as specified in 4.8.9.

3.3 Appearance. The rinse additive shall be distinctly colored with a stable colorant so that the liquid level in the rinse additive reservoir specified in MIL-I-43728 will be readily visible. After storage for 1 year in its original sealed packaging, the rinse additive shall still remain distinctly colored.

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3.4 Ash content. The rinse additive concentrate shall have an ash content not greater than 0.25 percent when tested as specified in 4.8.1.

3.5 Cloud point. The rinse additive concentrate shall have a cloud point not less than 60°F. (15.6°C.), and not greater than 120°F. (48.9°C.) when tested as specified in 4.8.2.

3.6 Surface tension. A 0.1 percent (by weight) solution of the rinse additive shall have a surface tension not greater than 45.0 dynes/cm., at 25°C., when tested as specified in 4.8.3.

3.7 Foam height. A 0.01 percent (by weight) solution of rinse additive shall have a foam height not greater than 1.5 centimeters (5-second value) when tested as specified in 4.8.4.

3.8 Solution stability. The rinse additive concentrate shall remain homogeneous and show no precipitation or phase separation when tested as specified in 4.8.5.

3.9 Nonvolatile matter. The rinse additive concentrate shall have a nonvolatile matter content not less than 40 percent when tested as specified in 4.8.6.

3.10 pH value. A 1.0 percent (by volume) distilled water solution of rinse additive shall have a pH value not less than 5.5 nor more than 8.5 when tested as specified in 4.8.7.

3.11 Biodegradability. The synthetic detergent in the rinse additive shall be 90 percent (minimum) biodegradable when tested as specified in 4.8.8.

3.12 Labeling. The containers shall be labeled in accordance with the Federal Hazardous Substances Act. Each container of rinse additive shall be durably and legibly marked with the following precautions and directions for use:

CAUTION: MAY BE HARMFUL IF SWALLOWED. Avoid contact with eyes. If concentrated liquid is splashed in eyes, flush thoroughly with water for at least 15 minutes and get medical attention. If swallowed, drink large amounts of water, induce vomiting, and call physician.

DIRECTIONS FOR USE: The supplier's recommended dilutions and "Directions for Use" shall be clearly shown on the label.

3.13 Workmanship. The rinse additive shall be clear, colored liquid, free from suspended or sedimentary matter.

3.14 Toxic products and formulations. This material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring agency (see 4.9).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the type, submitted for inspection and delivery at one time for examinations under 4.4 and 4.5.

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4.3 Inspection of material and components. In accordance with 4.1, the suppli is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications, drawings, and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.4 Examination of the end item for defects in appearance of rinse additive. The sample unit for this examination shall be one open, filled primary container. The inspection level shall be S-3 with an AQL of 2.5 expressed in terms of percent defective.

Examine	Defect
Appearance of rinse additives	Not clear. Not colored. Any sediment. Any suspended matter.

4.5 Examination of the end item for net contents. The sample unit for this examination shall be one filled container. The lot shall be unacceptable if the average net contents per container is less than specified or indicated for all the sample units examined. The inspection level shall be S-3.

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored as specified in table I below. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of percent defective.

TABLE I. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers; cushioning inadequate, improper, or missing.

4.6.1 Examination for closure, waterproofing, and banding of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix of that specification.

4.7 Testing of end item. The rinse additive shall be tested for the characteristics listed in table II. The lot size shall be the number of filled shipping containers in the end item inspection lot. Sixteen ounces of rinse additive shall be drawn from each of five shipping containers, selected at random, and composited. The test sample shall consist of 0.5 gallon (64 ounces) of the composite. The sample unit shall be placed in a clean, dry, glass container. Unless otherwise specified, duplicate determinations shall be performed on the composite with no evidence of failure to meet the specified requirements. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirements specified.

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TABLE II. Instructions for testing

Characteristic	Specification reference		Number determinations per unit composite	Results reported as	
	Requirement	Test method		Pass or fail	Numerically to nearest
Color stability	3.3	-	-	-	-
Ash content	3.4	4.8.1	Av. of 2	-	0.01 percent
Cloud point	3.5	4.8.2	Av. of 2	-	1.0 degree
Surface tension	3.6	4.8.3	Av. of 2	-	0.1 dyne/cm
Foam height	3.7	4.8.4	Av. of 2 at 82°C.	-	0.1 cm.
			Av. of 2 at 66°C.	-	0.1 cm.
Solution stability	3.8	4.8.5	1	X	-
Nonvolatile matter	3.9	4.8.6	Av. of 2	-	1.0 percent
pH value	3.10	4.8.7	Av. of 2	-	0.1 pH unit
Biodegradability (synthetic detergent)	3.11	4.8.8	2	-	1.0 percent
Ethylene glycol	3.2	4.8.9	2	X	-

4.8 Test methods.

4.8.1 Ash content. Ash content shall be determined in accordance with ASTM D 482.

4.8.2 Cloud point. Cloud point shall be determined in accordance with ASTM D 2024.

4.8.3 Surface tension. Surface tension shall be determined in accordance with ASTM D 1331.

4.8.4 Test for foam height.

4.8.4.1 Equipment.

(a) Stirring motor. The stirring motor shall be a standard Hamilton Beach No. 936 mixer or equal, with the stirring blade attached. The motor, with the stirring blade attached, shall be removed from its original stand and supported on a ring stand by means of a 250-ml beaker clamp.
(b) Test cell. The test cell shall be a glass cylinder 7.0 cm. inside diameter and 18.5 cm. high, made by sealing one end to provide a flat bottom.
(c) Baffle cage. The baffle cage shall be constructed as shown in figure I.
(d) Water bath. The water bath shall be used to control the temperature ± 0.5°C. (0.9°F.), in the test cell.

4.8.4.2 Procedure. Two hundred ml of 0.01 percent test solution of rinse additive shall be placed in the test cell and allowed to equilibrate in a 82° ± 0.5°C. (180° ± 0.9°F.) bath. The stirrer shall be centered between the four baffles and clamped in position so that the liquid level shall be midway between the two blades on the stirrer. The stirring motor shall be turned on and run for 3 minutes at medium speed (approximately 19,000 rpm) to allow the foam to reach maximum height. The stirrer shall be stopped after running 3 minutes ± 2 seconds, and foam height measured in centimeters after 5 seconds. The foam height shall be determined by measuring the total height in centimeters of the combined foam and liquid (from the bottom of the test cell), and subtracting the initial height of liquid measured prior to agitation. The difference is proportional to the volume of entrapped air and shall be designated as foam height. Distilled or deionized water shall be required for the above test. The test shall be rerun as above, using a 66°C. (150.8°F.) water bath.

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4.8.4.3 Reporting results. Two tests are run with each solution and the average of both 5-second results shall be the reported value. This shall be done for both temperature settings specified above, and both results shall be reported.

4.8.5 Solution stability. Accurately measure 200 ml of the rinse additive into a 200-ml tall form (Berzelius) beaker, cover with a watch glass, and place it in a constant temperature water bath or a controlled temperature oven at 180°F. (82°C.) for 3 hours. After removal from the oven or water bath, uncover the beaker and allow the rinse additive to attain room temperature 22.2 + 2°C (72° + 3°F.). Stir slowly for 1 minute and allow to stand for 15 minutes. There shall be no separating, layering, crystallizing, crust or film formation, or hardening of the surface of the sample.

4.8.6 Nonvolatile matter. Weigh to the nearest 0.001 g, 20 to 25 g of the rinse additive and place in a tared evaporating dish. Determine the nonvolatile matter in accordance with ASTM D 2834.

4.8.7 pH value. The pH shall be determined in accordance with ASTM E 70.

4.8.8 Biodegradability of the synthetic detergents.

4.8.8.1 Anionic synthetic detergents. When the detergent is an alkyl benzene sulfonate or a linear alkylate sulfonate, the biodegradability shall be determined in accordance with the Test Procedure and Standards, ABS and LAS Biodegradability Scientific and Technical Report No. 3 of the Soap and Detergent Association.

4.8.8.2 Nonionic and other synthetic detergents. The synthetic detergent shall be determined for biodegradability in accordance with the Status of Biodegradability Testing of Nonionic Surfactant Scientific and Technical Report No. 6 of the Soap and Detergent Association.

4.8.9 Test for ethylene glycol.

4.8.9.1 Reagents.

(a) 2.9 g sodium sulfanilate and 0.7 g sodium nitrite dissolved in 30 ml water
(b) 1.8 g α-napthylamine dissolved in 40 ml ethanol or dioxane.

4.8.9.2 Procedure. Several drops each of the reagents (a) and (b) are placed in a depression of a spot plate. A small amount of boric acid crystals are added and stirred with a glass rod. Then a drop of rinse additive is added. An immediate orange color indicates the presence of ethylene glycol.

4.9 Toxicological data and formulations. The contractor shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Rinse additive shall be packaged in 1-pint or 1-gallon capacity polyethylene bottles, as specified (see 6.2). Each polyethylene bottle shall be fabricated and closed in accordance with the applicable requirements of MIL-B-26701. The 1-gallon bottle shall be provided with a handle that is formed as an integral part of the bottle.

5.1.2 Level B. Rinse additive shall be packaged in 1-pint or 1-gallon quantities in accordance with the applicable requirements specified in 5.1.1 (see 6.2). Unless otherwise specified (see 6.2), the secondary closure or outer seal shall be a cellulose hydrate band, heat-shrinkable polyvinyl chloride tubing or equivalent, which covers the head of the container and extends to at least halfway up the sides of the screw cap. When dried, the band shall conform to the contour of the cap and neck of the bottle to prevent loosening of the cap. The band shall be treated to inhibit the growth of fungus. The outer seal shall not crack, loosen, or become dislodged.

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5.1.3 Level C. Rinse additive shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving agency. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve 1-pint or four 1-gallon capacity bottles of rinse additive, packaged as specified in 5.1.1, shall be packed in a fiberboard shipping container conforming to PPP-B-636, style RSC-L, grade V2s. In addition, each container shall be provided with top and bottom pads fabricated from the same material as the box liner. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each shipping container shall be closed, waterproofed, and banded in accordance with the appendix of the container specification.

5.2.2 Level B.

5.2.2.1 Military agencies. Twelve 1-pint or four 1-gallon capacity bottles of rinse additive, packaged as specified in 5.1.2, shall be packed in a fiberboard shipping container conforming to PPP-B-636, style RSC-L, class domestic, grade 275. In addition, each container shall be provided with top and bottom pads fabricated from the same material as the box liner. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1.1 When specified (see 6.2), the shipping container shall be a box conforming to PPP-B-636, grade V3c, V3s, or V4s, closed in accordance with the appendix to PPP-B-636.

5.2.2.2 Civil agencies. Twelve 1-pint or four 1-gallon bottles of rinse additive, packaged as specified in 5.1.2, shall be packed in a fiberboard shipping container conforming to PPP-B-636, class domestic, grade 275. Alternatively, the box shall be not less than grade 200, provided the basis weight of the corrugated medium used in the construction of the boxes shall be not less than 33 pounds per thousand square foot. Each box shall be fitted with a partition, divider, or liner, fabricated or fiberboard, of the same class and grade as the box. The box shall be 3/8 inch $\pm$ 1/8 inch higher than the filled closed bottles.

5.2.3 Level C. Rinse additive, packaged as specified in 5.1.3, shall be packed in a manner to insure carrier acceptance and safe delivery to destination. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Civil agencies. Marking of the unit and shipping containers shall be in accordance with 3.12 and Fed. Std. No. 123.

5.3.2 Military requirements. In addition to the labeling specified in 3.12, and any special marking required by the contract or order, marking of the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

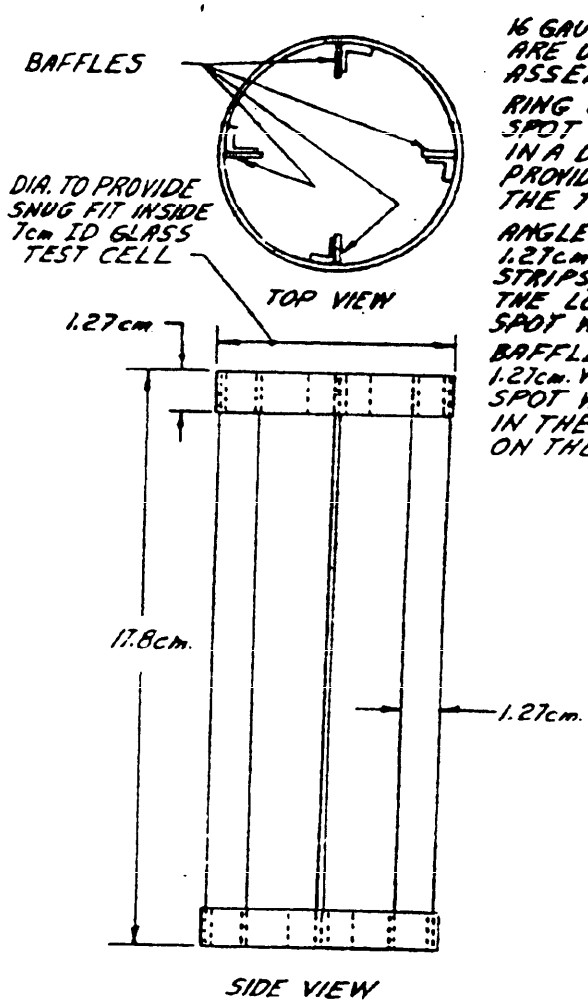
6. NOTES

6.1 Intended use. The liquid rinse additive concentrates are intended for use in the final rinse of mechanical dishwashing machines to aid in the elimination of water spotting and filming of tableware in water of any hardness.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and level of packaging and packing required (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2).
(d) Marking requirement (see 5.3).
(e) Quantity required (see 5.1).

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16 GAUGE STAINLESS STEEL STRIPS ARE USED THROUGHOUT THE ASSEMBLY.

RING JOINTS ARE LAPPED, SPOT WELDED AND FEATHERED IN A DIAMETER CAPABLE OF PROVIDING A SNUG FIT INSIDE THE 7cm ID GLASS TUBING.

ANGLES ARE FORMED BY USING 1.27cm WIDE BY 2.54cm LONG STRIPS, BENT IN THE CENTER OF THE LONG DIRECTION AND SPOT WELDED TO THE FORMED RINGS.

BAFFLES ARE FORMED USING 1.27cm. WIDE BY 17.8cm LG STRIPS, SPOT WELDED TO THE ANGLES IN THE POSITIONS INDICATED ON THE DIAGRAM.

FIG 1 BAFFLE CAGE FOR FOAM HEIGHT DETERMINATION

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MILITARY INTERESTS:

Custodians:

Army - GL (MCA)
Navy - MC
Air Force - 84

Review activities:

Army - MD
Air Force - 03

User activities:

Navy - SH, MS

CIVIL AGENCY COORDINATING ACTIVITIES

CPSC
GSA - FSS
HEW - NIH
USDA - ARS
VA - DMS

PREPARING ACTIVITY:

GSA - FSS

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